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Substrate conveying robot system

Abstract

A substrate conveying robot includes a hand sensor provided on a substrate holding hand, and a hand circuit substrate provided on the substrate holding hand and connected to the hand sensor, and a serial communication connection is made between a robot controller and the hand circuit substrate through an inside of an arm.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2015/0127147	12/2014	Yamazaki et al.	N/A	N/A
2016/0344266	12/2015	Ogawa et al.	N/A	N/A
2017/0165838	12/2016	Sato	N/A	N/A
2017/0246741	12/2016	Kobayashi	N/A	B25J 9/126
2020/0068739	12/2019	Tagashira	N/A	N/A
2020/0238535	12/2019	Tokushima et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2007203402	12/2006	JP	N/A
4466785	12/2009	JP	B25J 9/042
2012152843	12/2011	JP	N/A
2013-69914	12/2012	JP	N/A
2015-89583	12/2014	JP	N/A
2015089583	12/2014	JP	N/A
2017-104930	12/2016	JP	N/A
6165286	12/2016	JP	N/A
6299670	12/2017	JP	N/A
2018-089764	12/2017	JP	N/A
6365736	12/2017	JP	N/A
2018-142100	12/2017	JP	N/A
2020-121362	12/2019	JP	N/A
2019/014146	12/2018	WO	N/A

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a substrate conveying robot system, and more particularly, it relates to a substrate conveying robot system including a hand sensor provided on a substrate holding hand.

BACKGROUND ART

(2) Conventionally, a substrate conveying robot including a substrate conveying hand is known. Such a substrate conveying robot is disclosed in Japanese Patent Laid-Open No. 2013-069914, for example.

(3) Japanese Patent Laid-Open No. 2013-069914 discloses a substrate conveying robot that is a horizontal articulated robot. This substrate conveying robot includes a base, an arm connected to the base and rotating in a horizontal plane, and substrate conveying hands connected to the arm and rotating in the horizontal plane. A controller is connected to the substrate conveying robot. The controller controls the operation of starting and stopping the substrate conveying robot. Thus, substrates are conveyed by the substrate conveying robot.

(4) Although not clearly described in Japanese Patent Laid-Open No. 2013-069914, in a conventional substrate conveying robot as described in Japanese Patent Laid-Open No. 2013-069914, sensors may be provided on substrate conveying hands to detect substrates held by the substrate conveying hands. In this case, signals are transmitted from the sensors to a controller by wiring through the inside of an arm.

PRIOR ART

Patent Document

(5) Patent Document 1: Japanese Patent Laid-Open No. 2013-069914

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(6) In the aforementioned substrate conveying robot including the sensors, when each of the sensors communicates with the robot controller through the inside of the arm, a number of lines of wiring corresponding to the number of sensors are inserted from the substrate holding hands toward the controller through the inside of the arm. Therefore, the wiring becomes complex, and the inside of the arm needs to be relatively large. Thus, the size of the substrate conveying robot is disadvantageously increased.

(7) The present invention is intended to solve the above problems. The present invention aims to provide a substrate conveying robot system capable of reducing or preventing the complexity of wiring and an increase in the size of an arm (an increase in the size of a substrate conveying robot).

Means for Solving the Problems

(8) In order to attain the aforementioned object, a substrate conveying robot system according to an aspect of the present invention includes a substrate conveying robot, and a robot controller configured or programmed to control the substrate conveying robot. The substrate conveying robot includes an arm, a substrate holding hand moved by the arm, rotatably connected to the arm, and including a blade that supports a substrate, a hand sensor provided on the substrate holding hand, and a hand circuit substrate provided on the substrate holding hand and connected to the hand sensor. A serial communication connection is made between the robot controller and the hand circuit substrate through an inside of the arm.

(9) In the substrate conveying robot system according to this aspect of the present invention, as described above, a serial communication connection is made between the robot controller and the hand circuit substrate connected to the hand sensor through the inside of the arm. Accordingly, the hand sensor and the robot controller can communicate with each other by a relatively small number of lines of wiring for the serial communication connection between the robot controller and the hand circuit substrate. Even when a plurality of hand sensors are provided, the plurality of hand sensors are connected to the hand circuit substrate by a plurality of lines of wiring while the robot

controller and the hand circuit substrate communicate with each other by a relatively small number of lines of the wiring. Consequently, the complexity of the wiring and an increase in the size of the arm (an increase in the size of the substrate conveying robot) can be reduced or prevented due to a small number of lines of the wiring.

(10) Furthermore, the length of the wiring between the hand sensor and the hand circuit substrate is relatively short, and thus an increase in the influence of noise from the outside due to an increase in the length of the wiring can be reduced or prevented.

Effect of the Invention

(11) According to the present invention, as described above, it is possible to reduce or prevent the complexity of the wiring and an increase in the size of the arm (an increase in the size of the substrate conveying robot).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram showing the configuration of a substrate conveying robot according to an embodiment of the present invention.

(2) FIG. 2 is a schematic view of a substrate conveying robot system according to the embodiment of the present invention.

(3) FIG. 3 is a schematic view of a substrate conveying robot system according to a comparative example.

(4) FIG. 4 is a schematic view of a substrate conveying robot system according to a first modified example of the present invention.

(5) FIG. 5 is a schematic view of a substrate conveying robot system according to a second modified example of the present invention.

(6) FIG. 6 is a schematic view of a substrate conveying robot system according to a third modified example of the present invention.

(7) FIG. 7 is a schematic view of a substrate conveying robot system according to a fourth modified example of the present invention.

MODES FOR CARRYING OUT THE INVENTION

(8) An embodiment embodying the present invention is hereinafter described on the basis of the drawings.

(9) The configuration of a substrate conveying robot system **100** according to this embodiment is now described with reference to FIGS. 1 and 2.

(10) As shown in FIGS. 1 and 2, the substrate conveying robot system **100** according to this embodiment includes a substrate conveying robot **1** and a robot controller **2** (see FIG. 2) that controls the substrate conveying robot **1**. In the substrate conveying robot **1**, an arm **10** is moved up and down with respect to a substantially columnar housing **50** by an arm elevating mechanism (not shown) that moves the arm **10** up and down.

(11) The substrate conveying robot **1** is a horizontal articulated robot. Specifically, the substrate conveying robot **1** includes the arm **10**, a substrate holding hand **20** moved by the arm **10**, rotatably connected to the arm **10**, and including blades **21** that support substrates (semiconductor wafers) **W**, substrate detection sensors **22** provided on the substrate holding hand **20**, and a digital input circuit substrate **23** provided on the substrate holding hand **20** and connected to the substrate detection sensors **22**. The substrate detection sensors **22** are connected to the digital input circuit substrate **23** by wiring **23a**. The substrate holding hand **20** includes an upper substrate holding hand **20a** and a lower substrate holding hand **20b** arranged below the upper substrate holding hand **20a** and operating independently of the upper substrate holding hand **20a**. The wiring **23a** is separate from wiring **30** described below.

(12) In this embodiment, a plurality of blades **21** are provided, and a plurality of substrate detection sensors **22** are provided so as to correspond to the plurality of blades **21**. Each of the plurality of substrate detection sensors **22** is connected to the digital input circuit substrate **23** by the wiring **23a**. The plurality of blades **21** are spaced apart from each other along a vertical direction. The plurality of blades **21** include a plurality of blades **21a** attached to the upper substrate holding hand and a blade **21b** attached to the lower substrate holding hand **20b**.

(13) In this embodiment, a serial communication connection (see the wiring **30** in FIG. 2) is made between the robot controller **2** and the digital input circuit substrate **23** through the inside of the arm **10**. The digital input circuit substrate **23** is an example of a “hand circuit substrate” in the claims.

(14) In this embodiment, the substrate conveying robot system **100** includes an analog sensor **11** provided on the arm **10**, an analog input circuit substrate **12** provided on the arm **10** and connected to the analog sensor **11**, and a digital input/output circuit substrate **13** connected to the substrate detection sensors **22**. A serial communication connection is made between the robot controller **2**, the analog input circuit substrate **12**, the digital input/output circuit substrate **13**, and the digital input circuit substrate **23** through the inside of the arm **10** by daisy chain connection. The analog sensor **11** and the substrate detection sensors **22** are examples of an “arm sensor” and a “hand sensor” in the claims, respectively. The analog input circuit substrate **12** is an example of an “arm circuit substrate” or a “first arm circuit substrate” in the claims. The digital input/output circuit substrate **13** is an example of an “arm circuit substrate” or a “second arm circuit substrate” in the claims.

(15) In this embodiment, a serial communication connection is made between the robot controller **2**, the analog input circuit substrate **12**, and the digital input circuit substrate **23** through the inside of the arm **10** and movable portions **P1** at which the substrate holding hand **20** and the arm **10** are relatively movable by daisy chain connection. Inside a base **40** on which the arm **10** is relatively movably arranged, a movable portion **P2** is provided at which the wiring **30** for serial communication connection is movable. Tubular members (not shown) are provided in the movable portions **P1**, and the wiring **30** passes through the insides of the tubular members.

(16) In this embodiment, the arm **10** includes a first arm **10a** that is rotatable with a first end as the center of rotation, and a second arm **10b** including a first end rotatably connected to a second end of the first arm **10a**, rotatable with respect to the first arm **10a**, and connected to the substrate holding hand **20**. The analog input circuit substrate **12** and the digital input circuit substrate **23** are provided on at least one of the first arm **10a** or the second arm **10b**.

(17) As shown in FIG. 2, both the first arm **10a** and the second arm **10b** rotate along the horizontal plane. A space is provided inside the first arm **10a** and the second arm and the wiring **30** passes through the inside of the space.

(18) In this embodiment, the analog input circuit substrate **12** is provided on the first arm **10a**, and the digital input circuit substrate **23** is provided on the second arm **10b**. The analog input circuit substrate **12** is arranged in the space inside the first arm **10a**, and the digital input circuit substrate **23** is arranged in the space inside the second arm **10b**.

(19) In this embodiment, a signal is input from the analog sensor **11** to the analog input circuit substrate **12**. The analog sensor **11** is connected to the analog input circuit substrate **12** by wiring **12a**. The wiring **12a** is separate from the wiring **30a**.

(20) In this embodiment, the analog sensor **11** detects other than rotation of a motor provided at a joint of the arm **10**. Specifically, the analog sensor **11** includes at least one of a temperature sensor **11a**, an acceleration sensor **11b**, or a pressure sensor **11c**. In this embodiment, all of the temperature sensor **11a**, the acceleration sensor **11b**, and the pressure sensor **11c** are provided.

(21) In this embodiment, the digital input/output circuit substrate **13** is connected to one of the substrate detection sensors **22** and transmits and receives a signal to and from the substrate detection sensor **22**. The digital input/output circuit substrate **13** is connected to the substrate

detection sensor **22** by wiring **13a**. The wiring **13a** is separate from the wiring **30**.

(22) In this embodiment, the substrate conveying robot system **100** further includes electromagnetic valves **14** provided on the second arm **10b** to transmit and receive signals to and from the digital input/output circuit substrate **13** and adjusting air sent to cylinders **24** provided on the substrate holding hand **20**. The electromagnetic valves **14** are used to control opening and closing of pipes through which a fluid (such as air) passes. The digital input/output circuit substrate **13** is connected to the electromagnetic valves **14** by wiring **13a**.

(23) In this embodiment, the substrate detection sensors **22** detect other than rotation of the motor. Specifically, the substrate detection sensors **22** detect the presence of the substrates **W** supported by the blades **21**. The digital input circuit substrate **23** receives signals from the substrate detection sensors **22**. The substrate detection sensors **22** are any one of reflective optical sensors, transmissive optical sensors, capacitance sensors, distance sensors, and touch sensors designed on the assumption that they contact the substrates **W**. In this embodiment, the substrate detection sensors **22** are reflective optical sensors.

(24) In this embodiment, a serial communication connection is made between the robot controller **2** and the digital input circuit substrate **23** by a communication network that allows mutual information sharing between the robot controller **2** and the digital input circuit substrate **23**. Specifically, the robot controller **2** and the analog input circuit substrate **12** are connected to each other by the wiring **30** for serial communication connection. The analog input circuit substrate **12** and the digital input/output circuit substrate **13** are connected to each other by the wiring **30** for serial communication connection. The digital input/output circuit substrate **13** and the digital input circuit substrate **23** are connected to each other by the wiring **30** for serial communication connection. ICs are provided to establish a communication network that allows mutual information sharing on the analog input circuit substrate **12**, the digital input/output circuit substrate **13**, and the digital input circuit substrate **23**.

(25) Inside the base **40**, a power supply substrate **41** of an encoder (not shown) and an ID (identification) substrate **42** are provided.

(26) A substrate conveying robot system **600** according to a comparative example in FIG. **3** is now described.

(27) When a plurality of substrate detection sensors **22** and other sensors (such as an analog sensor **11** and electromagnetic valves **14**) communicate with a robot controller **2** through the inside of an arm **10** as in the substrate conveying robot system **600** according to the comparative example in FIG. **3**, a number of lines of wiring corresponding to the number of substrate detection sensors **22** and other sensors are inserted to the robot controller **2** side through the inside of the arm **10**. Therefore, it is necessary to relatively enlarge the inside of the arm **10**, and thus the size of a substrate conveying robot **1** is increased. In FIG. **3**, the lines of the wiring **30** extending from the plurality of substrate detection sensors **22** are grouped together in a substrate holding hand **20**, but in reality, a plurality of lines of the wiring **30** extending from the plurality of substrate detection sensors **22** are connected to the robot controller **2** through the inside of the arm **10**.

Advantages of This Embodiment

(28) According to this embodiment, the following advantages are achieved.

(29) According to this embodiment, as described above, a serial communication connection is made between the robot controller **2** and the digital input circuit substrate **23** connected to the substrate detection sensors **22** through the inside of the arm **10** such that the substrate detection sensors **22** and the robot controller **2** can communicate with each other by a relatively small number of lines of the wiring **30** for the serial communication connection between the robot controller **2** and the digital input circuit substrate **23**. Even when a plurality of substrate detection sensors **22** are provided, the plurality of substrate detection sensors **22** are connected to the digital input circuit substrate **23** by a plurality of lines of the wiring **30** while the robot controller **2** and the digital input circuit substrate **23** communicate with each other by a relatively small number of lines of the

wiring **30**. Consequently, the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented due to a small number of lines of the wiring **30**.

(30) Furthermore, the length of the wiring **30** between the substrate detection sensors **22** and the digital input circuit substrate **23** is relatively short, and thus an increase in the influence of noise from the outside due to an increase in the length of the wiring **30** can be reduced or prevented.

(31) According to this embodiment, as described above, the substrate conveying robot system **100** includes the analog sensor **11** provided on the arm **10**, the analog input circuit substrate **12** provided on the arm **10** and connected to the analog sensor **11**, and the digital input/output circuit substrate **13** connected to the substrate detection sensors **22**. A serial communication connection is made between the robot controller **2**, the analog input circuit substrate **12**, the digital input/output circuit substrate **13**, and the digital input circuit substrate **23** through the inside of the arm **10** by daisy chain connection. Accordingly, unlike a case in which the substrate detection sensors **22** and the analog sensor **11** are connected to the robot controller **2** by the wiring **30**, the substrate detection sensors **22** and the analog sensor **11** can communicate with the robot controller **2** by a relatively small number of lines of the wiring **30** for the serial communication connection between the robot controller **2**, the analog input circuit substrate **12**, the digital input/output circuit substrate **13**, and the digital input circuit substrate **23**. Consequently, in the configuration in which the analog input circuit substrate **12** and the digital input/output circuit substrate **13** are provided, the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(32) According to this embodiment, as described above, a serial communication connection is made between the robot controller **2**, the analog input circuit substrate **12**, the digital input/output circuit substrate **13**, and the digital input circuit substrate **23** through the inside of the arm and the movable portions **P1** at which the substrate holding hand **20** and the arm **10** are relatively movable by daisy chain connection. Accordingly, the robot controller **2**, the analog input circuit substrate **12**, the digital input/output circuit substrate **13**, and the digital input circuit substrate **23** are connected in series by daisy chain connection, and thus the number of lines of the wiring **30** through the movable portions **P1** is relatively small. Consequently, unlike a case in which a relatively large number of lines of the wiring **30** pass through the movable portions **P1**, the movable portions **P1** can be easily moved.

(33) According to this embodiment, as described above, the arm **10** includes the first arm **10a** rotatable with the first end as the center of rotation, and the second arm **10b** including the first end rotatably connected to the second end of the first arm **10a**, rotatable with respect to the first arm **10a**, and connected to the substrate holding hand and the analog input circuit substrate **12** and the digital input/output circuit substrate **13** are provided on at least one of the first arm **10a** or the second arm **10b**. Accordingly, in the substrate conveying robot system **100** including the analog input circuit substrate **12** and the digital input/output circuit substrate **13** on at least one of the first arm **10a** or the second arm **10b**, the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(34) According to this embodiment, as described above, the analog input circuit substrate **12** is provided on the first arm **10a**, and the digital input/output circuit substrate **13** is provided on the second arm **10b**. Accordingly, in the substrate conveying robot system **100** including both the analog input circuit substrate **12** and the digital input/output circuit substrate **13**, the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(35) According to this embodiment, as described above, the substrate conveying robot system **100** includes the analog input circuit substrate **12** to which a signal is input from the analog sensor **11**. Accordingly, unlike a case in which the substrate detection sensors **22** and the analog sensor **11** are connected to the robot controller **2** by the wiring the substrate detection sensors **22** and the analog

sensor **11** can communicate with the robot controller **2** while the complexity of the wiring **30** is reduced or prevented.

(36) According to this embodiment, as described above, the analog sensor **11** detects other than rotation of the motor provided at the joint of the arm **10**. Accordingly, even when the analog sensor **11** is provided separately from a sensor that detects rotation of the motor, a serial communication connection is made between the analog sensor **11** and the robot controller **2** through the analog input circuit substrate **12**, and thus the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(37) According to this embodiment, as described above, the analog sensor **11** includes at least one of the temperature sensor **11a**, the acceleration sensor **11b**, or the pressure sensor **11c**. Accordingly, even when at least one of the temperature sensor **11a**, the acceleration sensor **11b**, or the pressure sensor **11c** is provided, a serial communication connection is made between at least one of the temperature sensor **11a**, the acceleration sensor **11b**, or the pressure sensor **11c** and the robot controller **2** through the analog input circuit substrate **12**, and thus the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(38) According to this embodiment, as described above, the digital input/output circuit substrate **13** connected to the substrate detection sensors **22** to transmit and receive signals to and from the substrate detection sensors **22** is provided. Accordingly, even when the substrate detection sensors **22** are provided, a serial communication connection is made between the substrate detection sensors **22** and the robot controller **2** through the digital input/output circuit substrate **13**, and thus the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(39) According to this embodiment, as described above, the substrate conveying robot system **100** includes the electromagnetic valves **14** provided on the second arm **10b** to transmit and receive signals to and from the digital input/output circuit substrate **13** and adjust the air sent to the cylinders **24** provided on the substrate holding hand **20**. Accordingly, even when the electromagnetic valves **14** are provided on the second arm **10b**, a serial communication connection is made between the electromagnetic valves **14** and the robot controller **2** through the digital input/output circuit substrate **13** of the second arm **10b**, and thus the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(40) According to this embodiment, as described above, the substrate detection sensors **22** detect other than rotation of the motor. Accordingly, even when the substrate detection sensors **22** are provided separately from the sensor that detects rotation of the motor, a serial communication connection is made between the substrate detection sensors **22** and the robot controller **2** through the digital input circuit substrate **23**, and thus the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(41) According to this embodiment, as described above, the substrate detection sensors **22** are provided to detect the presence of the substrates **W** supported by the blades **21**, and the digital input circuit substrate **23** receives signals from the substrate detection sensors **22**. Accordingly, even when the substrate detection sensors **22** are provided separately from the sensor that detects rotation of the motor, a serial communication connection is made between the substrate detection sensors **22** and the robot controller **2** through the digital input circuit substrate **23**, and thus the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

(42) According to this embodiment, as described above, a serial communication connection is made between the robot controller **2** and the digital input circuit substrate **23** by the communication network that allows mutual information sharing between the robot controller **2** and the digital input

circuit substrate **23**. Accordingly, mutual information is shared between the robot controller **2** and the digital input circuit substrate **23**, and thus the substrate conveying robot system **100** can be easily controlled.

(43) According to this embodiment, as described above, the plurality of blades **21** are provided, and the plurality of substrate detection sensors **22** are provided so as to correspond to the plurality of blades **21**. When the plurality of substrate detection sensors **22** are provided, the number of lines of the wiring **30** extending from the substrate detection sensors **22** increases. Therefore, the plurality of lines of the wiring **30** extending from the plurality of substrate detection sensors **22** are connected to the digital input circuit substrate **23**, and a serial communication connection is made between the digital input circuit substrate **23** and the robot controller **2** through the inside of the arm **10** such that even when the plurality of substrate detection sensors **22** are provided, the complexity of the wiring **30** and an increase in the size of the arm **10** (an increase in the size of the substrate conveying robot **1**) can be reduced or prevented.

Modified Examples

(44) The embodiment disclosed this time must be considered as illustrative in all points and not restrictive. The scope of the present invention is not shown by the above description of the embodiment but by the scope of claims for patent, and all modifications (modified examples) within the meaning and scope equivalent to the scope of claims for patent are further included.

(45) For example, while the example in which the analog input circuit substrate **12** is provided on the first arm and the digital input/output circuit substrate **13** is provided on the second arm **10b** has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, as in a substrate conveying robot system **200** according to a first modified example shown in FIG. **4**, the analog input circuit substrate **12** and the digital input/output circuit substrate **13** may not be provided on the arm **10**. In this case, a serial communication connection is made between the digital input circuit substrate **23** and the robot controller **2**.

(46) While the example in which the analog input circuit substrate **12** is provided on the first arm **10a**, and the digital input/output circuit substrate **13** is provided on the second arm **10b** has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, as in a substrate conveying robot system **300** according to a second modified example shown in FIG. **5**, whereas the digital input/output circuit substrate **13** may be provided on the arm **10**, the analog input circuit substrate **12** may not be provided. In this case, a serial communication connection is made between the digital input circuit substrate **23**, the digital input/output circuit substrate **13**, and the robot controller **2**.

(47) While the example in which the analog input circuit substrate **12** is provided on the first arm **10a**, and the digital input/output circuit substrate **13** is provided on the second arm **10b** has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, as in a substrate conveying robot system **400** according to a third modified example shown in FIG. **6**, whereas the analog input circuit substrate **12** may be provided on the arm **10**, the digital input/output circuit substrate **13** may not be provided. In this case, a serial communication connection is made between the digital input circuit substrate **23**, the analog input circuit substrate **12**, and the robot controller **2**.

(48) While the example in which the analog input circuit substrate **12** is provided on the first arm **10a**, and the digital input/output circuit substrate **13** is provided on the second arm **10b** has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, as in a substrate conveying robot system **500** according to a fourth modified example shown in FIG. **7**, both the analog input circuit substrate **12** and the digital input/output circuit substrate **13** may be provided on the second arm **10b**. Alternatively, both the analog input circuit substrate **12** and the digital input/output circuit substrate **13** may be provided on the first arm **10a**.

(49) While the example in which the plurality of blades **21** and the plurality of substrate detection sensors **22** are provided has been shown in the aforementioned embodiment, the present invention

is not limited to this. For example, the present invention may be applied to a substrate conveying robot system in which one blade **21** and one substrate detection sensor **22** are provided.

(50) While the example in which the arm **10** includes two arm portions, which are the first arm **10a** and the second arm **10b**, has been shown in the aforementioned embodiment, the present invention is not limited to this. For example, the number of arm portions of the arm has may be other than two.

(51) While the example in which the analog sensor **11** includes the temperature sensor **11a**, the acceleration sensor **11b**, and the pressure sensor **11c** has been shown in the aforementioned embodiment, the present invention is not limited to this. In the present invention, the analog sensor **11** may include sensors other than these sensors.

(52) While the example in which the “hand sensor” according to the present invention is the substrate detection sensor **22** has been shown in the aforementioned embodiment, the present invention is not limited to this. A sensor other than the substrate detection sensor **22** may be applied as the “hand sensor” according to the present invention.

DESCRIPTION OF REFERENCE NUMERALS

(53) **1**: substrate conveying robot **2**: robot controller **10**: arm **10a**: first arm **10b**: second arm **11**: analog sensor (arm sensor) **11a**: temperature sensor **11b**: acceleration sensor **11c**: pressure sensor **12**: analog input circuit substrate (arm circuit substrate, first arm circuit substrate) **13**: digital input/output circuit substrate (arm circuit substrate, second arm circuit substrate) **14**: electromagnetic valve **20**: substrate holding hand **21**: blade **22**: substrate detection sensor (hand sensor) **23**: digital input circuit substrate (hand circuit substrate) **24**: cylinder **P1**: movable portion **W**: substrate

Claims

1. A substrate conveying robot system comprising: a substrate conveying robot; and a robot controller configured or programmed to control the substrate conveying robot; wherein the substrate conveying robot includes: an arm; a substrate holding hand configured to be moved by the arm, rotatably connected to the arm, and including a plurality of blades, each blade is configured to support a separate corresponding substrate; a plurality of substrate detection sensors provided on the substrate holding hand, each of the substrate detection sensors corresponding to one of the plurality of blades and configured to detect a presence of the separate corresponding substrate on a corresponding blade; and a hand circuit substrate provided on the substrate holding hand and connected in parallel to each of the plurality of substrate detection sensors; and a serial communication connection is made between the robot controller and the hand circuit substrate through an inside of the arm, wherein the hand circuit substrate combines signals from the plurality of substrate detection sensors connected in parallel into the serial communication connection.
2. The substrate conveying robot system according to claim 1, further comprising: an arm sensor provided on the arm; and an arm circuit substrate provided on the arm and connected to at least one of the arm sensor or the plurality of substrate detection sensors; wherein the serial communication connection is made between the robot controller, the arm circuit substrate, and the hand circuit substrate through the inside of the arm by a daisy chain connection.
3. The substrate conveying robot system according to claim 2, wherein the serial communication connection is made between the robot controller, the arm circuit substrate, and the hand circuit substrate through the inside of the arm and through a rotatable joint with a passage for cabling at which the substrate holding hand and the arm are relatively movable with the daisy chain connection.
4. The substrate conveying robot system according to claim 2, wherein the arm includes a first arm rotatable with a first end as a center of rotation, and a second arm including a first end rotatably connected to a second end of the first arm, rotatable with respect to the first arm, and connected to

the substrate holding hand; and the arm circuit substrate is provided on at least one of the first arm or the second arm.

5. The substrate conveying robot system according to claim 4, wherein the arm circuit substrate includes a first arm circuit substrate provided on the first arm, and a second arm circuit substrate provided on the second arm.

6. The substrate conveying robot system according to claim 5, wherein the arm sensor includes an analog sensor; and at least one of the first arm circuit substrate or the second arm circuit substrate includes an analog input circuit substrate to which a signal is input from the analog sensor.

7. The substrate conveying robot system according to claim 6, wherein the analog sensor detects other than rotation of a motor provided at a joint of the arm.

8. The substrate conveying robot system according to claim 7, wherein the analog sensor includes at least one of a temperature sensor, an acceleration sensor, or a pressure sensor.

9. The substrate conveying robot system according to claim 5, wherein the second arm circuit substrate includes a digital input/output circuit substrate connected to the plurality of substrate detection sensors to transmit and receive a signal to and from the plurality of substrate detection sensors.

10. The substrate conveying robot system according to claim 9, further comprising: an electromagnetic valve provided on the second arm to transmit and receive a signal to and from the digital input/output circuit substrate and adjust air sent to a cylinder provided on the substrate holding hand.

11. The substrate conveying robot system according to claim 1, wherein the plurality of substrate detection sensors detects other than rotation of a motor.

12. The substrate conveying robot system according to claim 11, wherein the hand circuit substrate includes a digital input circuit substrate to receive a signal from the plurality of substrate detection sensors.

13. The substrate conveying robot system according to claim 1, wherein the serial communication connection allows mutual information sharing between the robot controller and the hand circuit substrate.
